

CS 472/572: Object-Oriented Design Patterns

Instructor

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Meetings

Lectures: MoWe 10:30–11:45 CCP-260
Office hours: MoWe 4:15–5:15 CCP-258
or by appointment CCP-258

Lectures will be audio/video recorded, and made available, afterwards.

I am happy to answer questions by email. Please please click [here](#), or see:

[pub/doc/EmailQuestions.pdf](#)

Catalog Description

Reviews object-oriented design principles, explains the goals and form of design patterns, and examines several well-known patterns.

PREREQ: CS 321.

Goals

The student will be able to explain and employ the basic concepts of object-oriented design patterns:

- understand the meaning and benefits of software reusability
- know the form of a software design pattern

- understand domain-independent versus domain-specific patterns
- understand how to apply a pattern
- identify relationships between patterns
- recognize a taxonomy of several well-known object-oriented design patterns
- understand class patterns versus object patterns
- recognize the value of programming to an interface
- understand class versus interface inheritance
- understand inheritance versus composition
- understand delegation
- understand inheritance versus parameterized types
- distinguish between run-time versus compile-time structures
- perform object-oriented design and programming in C++ and Java

Textbook

- *Design Patterns*, Erich Gamma, Richard Helm, Ralph Johnson, and John Vlissides, First edition, Addison-Wesley, 1995, ISBN: 9780201633610.

Other Course Material

This syllabus, lecture slides, assignments, and other material is available in what we'll called our “**pub**” directory. It is available in three places: GitHub, Canvas, and the **onyx** cluster of computers. This directory is read-only. So, you might want to copy it, perhaps to your local computer.

The GitHub **pub** directory can be accessed at [GitHub](#), as shown in:

[pub/GitHub](#)

The Canvas **pub** directory can be accessed from our [Canvas](#) website, via the “Files” tab on the left sidebar.

The **onyx** **pub** directory can be accessed directly, by computers in our Computer Science Lab (CCP-240, CCP-241, and CCP-242).

Since `onyx.boisestate.edu` services Secure Shell (SSH) requests, you can also use SSH clients (e.g., `scp` and `sftp`) to access this `pub` directory remotely. However, beware: It contains symbolic links to parent directories, and `scp -r` will unconditionally follow them, thereby looping forever. To avoid this, use `sftp` or `tar/scp`, as needed.

In any case, the `onyx pub` directory is at:

```
~jbuffenb/classes/472/pub
```

The `onyx` cluster also has the advantage of containing all of the translators we will use.

Grades

At the end of the semester, the Registrar requires a letter grade for each student. Accordingly, during the semester, homework and exams are evaluated, and numeric scores are assigned. Each such artifact is worth a certain number of points, and has a weight. From these scores, a student's overall numeric/raw percentage is computed.

Collectively, a class's raw percentages form a distribution, with a mean and standard deviation, sometimes called a Bell Curve. An algorithm normalizes these values into a "grading" distribution, with a desired mean and standard deviation of 85 and 10 (i.e., the values are "curved"). A normalized value is then mapped to a letter grade, in the conventional way. If you are interested in the gory details, click [here](#).

During the semester all of this evolving data can be found in our `pub` directory, on `onyx`, its nodes, Canvas, and GitHub. Homework is due at 11:59PM, Mountain Time, on the day it is due. Late work is not accepted. To submit your solution to an assignment, login to a lab computer, change to the directory containing the files you want to submit, and execute:

```
submit jbuffenb class assignment
```

For example:

```
submit jbuffenb cs472 hw1
```

The `submit` program has a nice `man` page.

When you submit a program, include: the source code, sample input data, and its corresponding results.

Scores are posted in our `pub/scores` directory, as they become available. You will receive a code, by email, indicating your row in the score sheet. You are encouraged to check your scores to ensure they are recorded properly. If you feel that a grading mistake has been made, contact me as soon as possible.

At the end of the course, a letter grade is assigned to each student according to rank among classmates, which is determined from numerical scores assigned for performance of these activities:

<i>Activity</i>	<i>Weight</i>
Homework	50%
Exam	25%
Final	25%

Homework

Several homework problems are assigned during the semester. Each asks you to develop software in the Java programming language. Students work on these individually.

The rubric that will be used to grade each assignment is distributed with the assignment. Try to focus on the assignments, rather than the rubrics.

Exams

An exam and a final are administered. These are in-class, open-note, open-textbook, and open-laptop tests. Students work on these individually.

Graduate-Student Grades

This is a dual-listed section: containing undergraduate and graduate students.

Graduate students are assigned extra work: additional assignments, or additional parts of existing assignments. It may take the form of software development, or research and in-class presentation. Details are determined by the number of graduate students enrolled. In any event, rubrics will reflect this extra work.

For graduate students, the “grading” distribution, discussed above, is different. It has a mean and standard deviation of 80 and 10. The effect is that a graduate student is expected to “do better” than an undergraduate student.

Attendance

In-person lecture attendance is an important part of course participation. Attendance is taken at each lecture: starting five minutes before the scheduled start time, and ending fifteen minutes after the scheduled start time. Attendance is not taken during the first week of classes, holidays, or finals week.

Attendance can affect your grade. Each absence results in a one-percent reduction of your overall normalized percentage. Since a few absences are expected, completion of BSU's on-line end-of-semester course evaluation will erase up to five absences. Since evaluations are anonymous (at least to me), I will try to remind you to email me evidence of completing one: a screenshot will do.

Attendance is administered wirelessly, via the iClicker app, available for free, from your smartphone's app store. For more information, click [here](#).

Source-Code Documentation

Good documentation and programming style is very important. Your programs must demonstrate these qualities for full credit. Good documentation and programming style includes:

- heading comments giving: author, date, class, and description
- function/procedure comments giving description of: purpose, parameters, and return value
- other comments where clarification of source code is needed
- proper and consistent indentation
- proper structure and modularity

For more information, and examples, click [here](#).

Academic Integrity

The University's goal is to foster an intellectual atmosphere that produces educated, literate people. Because cheating and plagiarism are at odds with that goal, those actions shall not be tolerated in any form. Academic dishonesty includes assisting a student to cheat, plagiarize, or commit any act of academic dishonesty. Plagiarism occurs when a person tries to represent another person's

work as his or her own or borrows directly from another person's work without proper documentation.

If a student engages in academic dishonesty, the student may be dismissed from the class and may receive a failing grade. Other penalties may include suspension or expulsion from the University.

For much more information about academic integrity, including examples of academic dishonesty, please click [here](#). If you are unsure about a particular behavior, ask your instructor.

Labs and Safety

Each student receives an account on the cluster of computers in the Computer Science Labs: CCP-240, CCP-241, and CCP-242. The cluster comprises a server named `onyx.boisestate.edu` and a set of nodes with shared home directories. It is remotely accessible, via SSH. The cluster runs the Linux operating system.

Physical access requires building and room access. After-hours building access, and all-hours room access, require an authenticated proximity-type student-identification card.

You are responsible for understanding and obeying lab [rules](#).

Schedule

<i>Week</i>	<i>Date</i>	<i>Topic</i>	<i>Assigned</i>	<i>Due</i>	<i>Reading</i>
1	Jan 12 Mon	Introduction			
	Jan 14 Wed				1
2	Jan 19 Mon	MLK Day			
	Jan 21 Wed				
3	Jan 26 Mon		HW1		
	Jan 28 Wed				
4	Feb 02 Mon	A Case Study: Designing a Document Editor			2ff
	Feb 04 Wed				
5	Feb 09 Mon				
	Feb 11 Wed				
6	Feb 16 Mon	Presidents' Day			
	Feb 18 Wed				
7	Feb 23 Mon				
	Feb 25 Wed				
8	Mar 02 Mon		HW2	HW1	
	Mar 04 Wed				
9	Mar 09 Mon				
	Mar 11 Wed	Exam			
10	Mar 16 Mon	Spring Break			
	Mar 18 Wed	Spring Break			
11	Mar 23 Mon		HW3	HW2	
	Mar 25 Wed				
12	Mar 30 Mon				
	Apr 01 Wed		HW4		
13	Apr 06 Mon			HW3	
	Apr 08 Wed				
14	Apr 13 Mon		HW5,HWG		
	Apr 15 Wed			HW4	
15	Apr 20 Mon				
	Apr 22 Wed				
16	Apr 27 Mon			HW5,HWG	
	Apr 29 Wed	HWG Presentations			
17	May 04 Mon	Final: 10:00			